Quantum N-gram Language Models for Tweet Classification

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***Abstract*—This paper presents a novel approach to text classification using n-grams and quantum machine learning models. We demonstrate the effectiveness of our approach by applying it to classify tweets, a challenging and noisy text classification task. Specifically, we show that using quantum models can significantly improve classification performance compared to classical ones. Additionally, we leverage quantum hardware to compute higher-order n-grams (more than bi-grams) and observe a boost in computation performance. Our experimental results demonstrate the potential of quantum computing for text classification tasks and pave the way for further exploration of quantum machine learning methods in natural language processing.**

***Index Terms*—quantum machine learning, natural language processing, n-grams, quantum computing**

I. Introduction

Text classification is a fundamental task in natural language processing (NLP) and has a wide range of applications in areas such as sentiment analysis, spam filtering, and topic modeling. The effectiveness of text classification algorithms largely depends on the features used to represent the text data. N-grams are a popular feature representation method in NLP, where N-grams are contiguous sequences of N items (e.g., words, characters) extracted from the text. The frequency of occurrence of N-grams in the text serves as a feature vector for classification.

There has been significant research on using classical machine learning algorithms for text classification tasks. Traditional approaches rely on bag-of-words representations, which suffer from high dimensionality and sparsity issues. N-grams are effective in capturing local context and improving classification performance. However, the computation of higher-order N-grams can be computationally expensive, and their effectiveness in improving classification performance is limited.

Quantum machine learning has also gained attention in recent years due to its potential to speed up certain computations by exploiting quantum phenomena such as superposition and entanglement. Several studies have explored using quantum algorithms for machine learning tasks, such as support vector machines, clustering, and principal component analysis [1]–[7]. However, the application of quantum machine learning to NLP is limited.

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One study proposed a novel approach to address privacy-preservation issues for speech recognition in federated learning. The proposed approach utilizes a quantum convolutional neural network (QCNN) composed of a quantum circuit encoder for feature extraction and a recurrent neural network (RNN) based end-to-end acoustic model (AM) to extract Mel-spectrogram and corresponding convolutional features. This approach enhances model parameter protection in a decentralized architecture by utilizing quantum learning progress to secure models and avoid privacy leakage attacks. In their experiments on the Google Speech Commands Dataset, the proposed QCNN encoder outperforms previous architectures using centralized RNN models with convolutional features [8]. On the other hand, C. -H. Huck Yang *et al.* propose a vertical federated learning architecture based on variational quantum circuits to demonstrate the competitive performance of a quantum-enhanced pre-trained BERT model for text classification. The proposed hybrid classical-quantum model consists of a novel random quantum temporal convolution (QTC) learning framework replacing some layers in the BERT-based decoder.

The experiments conducted on intent classification using the Snips and ATIS spoken language datasets show that the proposed BERT-QTC model achieves competitive experimental results. The results also demonstrate that the BERT-QTC model boosts the performance of the existing quantum circuit-based language model in two text classification datasets by 1.57% and 1.52% relative improvements [9].

Our work differs from these studies because we use quantum machine learning models for N-gram-based text classification. We show that quantum models can significantly improve classification performance compared to classical ones and that quantum hardware can improve computation performance for higher-order N-grams.

II. N-gram Language Models

N-gram Language Models are a type of predictive model that assigns probabilities to sequences of words. An N-gram is a sequence of N words, such as a bi-gram (two-word sequence) or a tri-gram (three-word sequence). These models can be used to estimate the probability of the last word in an N-gram given the previous words and assign probabilities to entire sequences. Efficiency considerations are essential when building language models that use large sets of N-grams.

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Rather than storing each word as a string, it is generally represented in memory as a 64-bit hash number, with the words themselves stored on disk. Probabilities are generally quantized using only 4-8 bits (instead of 8-byte floats), and N-grams are stored in reverse tries.

N-grams can provide valuable insights into the structure and meaning of the text, including the frequency of word sequences, the distribution of words within a text, and the relationships between words. As a result, they are helpful for various applications, including language modeling, text classification, machine translation, and information retrieval. One important consideration when working with N-grams is the potential for data sparsity, mainly when using higher values of N. It can be mitigated through techniques such as smoothing and backoff and using more advanced language models such as neural networks.

The complexity of an N-gram Language Model can be measured using perplexity, defined as two raised to the power of the cross-entropy of the model on a sequence of words. The cross-entropy of a model M on a sequence of words W is close to $H(W) = -1/NlogP(w1w2...wN)$, where $P(w1w2...wN)$ is the probability assigned by the model to the entire sequence. Perplexity measures how well the model predicts a test set, with lower values indicating better performance. However, as N increases, so does the number of possible N-grams and, thus, the size of the model's vocabulary and storage requirements. Therefore, a trade-off exists between model complexity and performance [10]–[12].

III. Quantum Natural Language Processing

Quantum natural language processing (QNLP) is an emerging field that combines the principles of quantum mechanics with natural language processing (NLP) to improve the processing of natural language data. QNLP promises to revolutionize the field of NLP by providing faster and more efficient algorithms for processing and analyzing natural language data.

Recent advancements in quantum computing and quantum algorithms have paved the way for developing QNLP applications. However, the field is still in its early stages, and much research is needed to develop practical QNLP applications that can outperform classical NLP systems.

One of the most promising applications of QNLP is in language modeling, which involves predicting the likelihood of a sequence of words given a preceding context. It is achievable using N-gram language models like the ones presented in this work. Recent research has shown that quantum language models have the potential to outperform classical language models on some specific tasks. However, quantum language models are still not competitive with classical models on most NLP tasks [13]–[17].

A. Our Approach

We developed three quantum-enhanced models and compared them in terms of performance to a classical support vector machine (C-SVM). To implement and evaluate these models, we used a combination of PennyLane, an open-source software framework for optimization and machine learning of quantum and hybrid quantum-classical computations [18] and Qiskit, another open-source software framework for the development of applications in quantum computing [19]. Specifically, we utilized the quantum simulators provided by PennyLane to develop our models. Also, we used an actual quantum processing unit (QPU) provided by IBM Quantum through Qiskit. This IBMQ-manila device has five qubits to execute our circuits on a real quantum computer.

To further boost the computational performance, we also employed a V100 NVIDIA GPU to parallelize the evaluations of the circuits using PennyLane. It allowed us to speed up the training process and reduce the time required to evaluate the models.

The three models that we developed are as follows: Quantum Kernel Support Vector Machine (QK-SVM), Variational Quantum Circuit or Quantum Neural Network (QNN), and Hybrid Quantum Convolutional Neural Network (H-QCNN).

*1) **Quantum Kernel Support Vector Machine (QK-SVM)**:* Researchers have made significant strides in quantum machine learning over the past few years in understanding and training quantum circuits as machine learning models. Although there have been many parallels to neural networks, it has become increasingly apparent that their underlying mathematics is closely associated with kernel methods. The support vector machine (SVM) is the most prominent example [20]–[22]. Our model uses a quantum circuit to encode the input data into a feature space, which serves to construct a kernel matrix. The kernel matrix trains a support vector machine (SVM) on the data. In order to implement the kernel, we first prepare the state vector $|\phi(x)\rangle$ and then apply the inverse embedding using $x^{'}$ on the same set of qubits. Subsequently, we measure the projector onto the initial state $|0..0\rangle\langle 0..0|$. We can confirm that this procedure yields the kernel as follows:

$$\begin{aligned} &\langle 0..0|S(x')S(x)^{\dagger}\mathcal{M}S(x')^{\dagger}S(x)|0..0\rangle = \\ &\langle 0..0|S(x')S(x)^{\dagger}|0..0\rangle\langle 0..0|S(x')^{\dagger}S(x)|0..0\rangle \\ &= |\langle 0..0|S(x')^{\dagger}S(x)|0..0\rangle|^2 \\ &= |\langle \phi(x')|\phi(x)\rangle|^2 \\ &= \kappa(x,x'). \end{aligned} \tag{1}$$

Thus, we can see that preparing the state vector and applying the inverse embedding to the same set of qubits leads to the kernel, which it can verify through Equation 1.

*2) **Variational Quantum Circuit or Quantum Neural Network (QNN)**:* In contemporary research, the term "quantum neural network" has been used to describe parametrized or variational quantum circuits. Although the mathematical underpinnings of these circuits differ substantially from traditional neural networks, the comparison emphasizes the modular aspect of quantum gates in a circuit and the frequent implementation of techniques from neural network training

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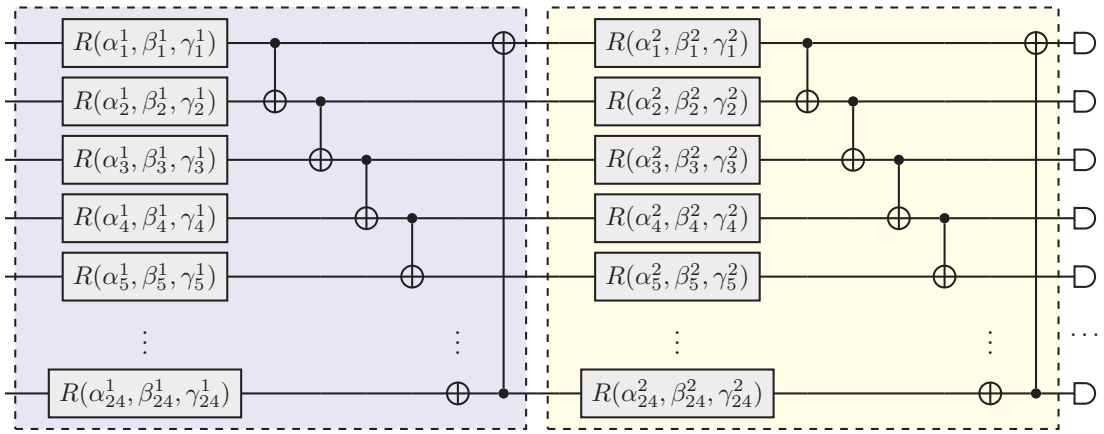


Figure 1. Variational quantum circuit layer. It consists of a quantum embedding layer that applies an angle embedding to the input, followed by several layers of entangler gates. The final output of the quantum layer is a sequence of expectation values of Pauli-Z operators.

to optimize quantum algorithms. This analogy has been instrumental in elucidating the relationship between these two seemingly distinct fields of study and has opened up new avenues of research in quantum computing. This model uses a variational quantum circuit to perform a classification task. During training, the circuit uses a gradient-based optimization algorithm to minimize the classification error. As is shown in Figure 1, it consists of a fixed series of layers (from two to seven in our implementation) conformed by entangling gates.

Based on the circuit-centric classifier design [23]. We used up to 25 qubits for this model.

3) ***Hybrid Quantum Convolutional Neural Network (H-QCNN)****:* This model combines classical and quantum components to perform a classification task. The classical component consists of a convolutional neural network (CNN), while the quantum component is a variational quantum circuit. The two components train together using a joint optimization algorithm. Figure 2 explains the architecture implemented for this model.

Overall, we evaluated the performance of these three models against a C-SVM, considering the N-gram used in each training process.

B. The Problem

The problem of classifying tweets is a relevant and challenging task in the field of natural language processing. Tweets are short and unstructured, making them difficult to classify accurately using traditional machine learning models such as logistic regression and support vector machines. It is because tweets contain high-dimensional, sparse, and noisy features that can be difficult to extract and process.

Quantum machine learning algorithms offer a promising approach to address these challenges. By leveraging the quantum nature of information processing, these models can efficiently process high-dimensional data and capture subtle patterns and correlations that classical models may miss. Additionally, quantum machine learning models such as quantum neural networks and quantum support vector machines outperform their classical counterparts in various tasks, including classification, regression, and clustering.

In particular, the problem of classifying tweets is relevant to quantum machine learning and quantum natural language processing because it poses unique challenges that quantum algorithms and models can address. With this, we explore new ways of applying quantum computing to real-world problems in social media analysis, such as sentiment analysis.

C. The Data

To implement our experiments, we use the SemEval-2018 Affect in Tweets Distanced Supervision Corpus [24]. We chose this dataset as it contains many tweets, which allows for robust training and testing of tweet classification models. Secondly, the dataset includes tweets that cover a wide range of topics and come from different geographic regions, which makes it more representative of the diverse nature of tweets. Finally, we carried out a pre-processing process for the data to be suitable for our implementation. As shown in Table I, we used the polarity of the emotion as the label for tweets written in Spanish only.

As a part of our implementation, we utilized various pre-processing techniques to clean and normalize the raw tweet data before inputting it into the quantum N-gram language model.

To identify the most frequently used words in the dataset, we conducted a word frequency count by iterating through each tweet and tokenizing it with the spaCy library. Then, we added only those words to a dictionary that was alphabetic and not stop words, along with their frequency count.

Additionally, we applied several pre-processing steps to each tweet. Firstly, we normalized the tweet text using Unicode normalization and ASCII encoding. Subsequently, we replaced emojis, URLs, mentions, and hashtags with unique

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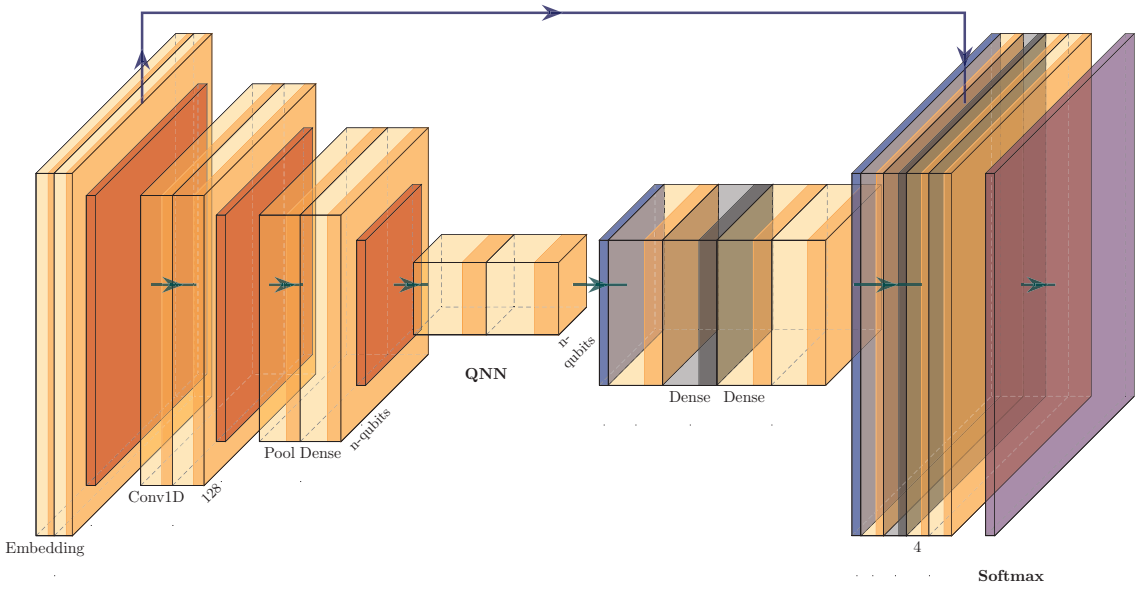


Figure 2. Hybrid quantum convolutional neural network architecture for tweet classification. The model consists of an embedding layer, a convolutional layer, and a quantum layer. The input is a sequence of n-features-dimensional vectors, first passed through an embedding layer to obtain a sequence of 128-dimensional vectors. Then, a convolutional layer with 128 filters and a kernel size of 2 applies a sliding window to extract local features, followed by a global max pooling layer. The output feeds into a quantum node that applies angle embedding and entangler gates. The final output is a sequence of expectation values of Pauli-Z operators, which are passed through a fully connected layer and a softmax activation function to obtain a probability distribution over the classes.

Table I
THE DATASET.

tweet id	user	content	date	lang	polarity
770976639173951488	noseashetero	@noseashetero 1000/10 de verdad a ti que voy a ...	31/08/2016 13:28	es	P
771092421866389508	Templelx	@piscolabisaereo @HistoriaNG @SPosteguillo las ...	31/08/2016 21:08	es	P
771092111429083136	esskuu94	Al final han sido 3h Bueno, mañana tengo fiest...	31/08/2016 21:07	es	P
771092070572449796	__ariadna9	@Jorge_Ruiz14 yo no tengo tiempo para esas cosa...	31/08/2016 21:07	es	N
771094192508600320	_cristtina15_	@_MissChaotic_ ves ese brillo? es un coso que h...	31/08/2016 21:15	es	N
771116956518875137	JSorgex	Tengo una perrina adorable... Sabéis que me aco...	31/08/2016 22:46	es	P
771115324884262912	Escarolilla	@juankipua Es que en el Ojeando el año pasado t...	31/08/2016 22:39	es	NEU
771118683414560768	KaichiZick95	Bueno, estoy en la batalla final del Conquista ...	31/08/2016 22:53	es	NONE
770550748107202560	carlosnmr	@CondeDuqueMAD ¿mañana sábado 31? En que día vi...	30/08/2016 9:16	es	NONE
770549862500802560	hermesni	@agcasti y el caminante ante el mar de niebla, ...	30/08/2016 9:12	es	P

tokens (”EMOJI”, ”URL”, ”MENTION”, and ”HASHTAG”, respectively) to eliminate noise from the text.

We also removed various patterns and symbols, including copyright symbols, brackets, punctuation marks, numbers, extra white spaces, and trailing spaces. Finally, we returned the pre-processed tweet text if it was not empty or just a space. Next, we represented each pre-processed tweet as a vector of word frequencies using the Bag-of-Words (BoW) approach, a prevalent natural language processing technique. This approach enabled us to convert text into a numerical format that could serve as input for machine learning algorithms. Finally, we employed N-grams of length two to nine to capture longer phrases and word combinations in our implementation.

IV. RESULTS

The results of this work are summarized in Table II and Table III. On that basis, we can point out a few things. From Table II, we can see that the accuracy of all models improves with an increase in N-gram size. For instance, the QK-SVM model achieves an accuracy of 0.3347 for 2-gram, which improves to 0.34614 and 0.41477 for 5-gram and 9-gram, respectively. On the other hand, the H-QCNN model has the highest accuracy of 0.375 for 5-gram, but its accuracy drops to 0.3043 for 2-gram and improves to 0.41825 for 9-gram.

From Table III, we can see that the H-QCNN model achieves the highest accuracy, followed by the QK-SVM, QNN, and C-SVM being the classical method, the one with the poorest performance. We evidence an improvement of

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Table II
ACCURACY OF MODELS VS. N-GRAM.

N-gram	C-SVM	QNN	QK-SVM	H-QCNN
2-gram	0.2045	0.24714	0.3347	0.3043
5-gram	0.2876	0.25	0.34614	0.375
9-gram	0.3125	0.3977	0.41477	0.41825

approximately 10% from the classical to the best quantum-enhanced method. However, the precision of all models is relatively low, indicating a high false positive rate. The recall and F-score of the models are also relatively low, indicating that the models have difficulty identifying all relevant tweets. While evaluating the models, we also compared the training execution time for CPU, GPU, and QPU when increasing the N-gram. We observe how the larger the N-gram, the task becomes computationally demanding due to the high dimensionality of the features. In this respect, we note that the QPU was the most consistent execution time, although being out-performed for low order N-grams evaluations, it presents the slightest variation when augmenting the order of the N-grams.

Table III
BEST PERFORMANCE BY MODEL.

Model	Accuracy	Recall	Precision	F-score	Missclass
C-SVM	0.3125	0.3125	0.58461	0.26912	0.6875
QNN	0.39772	0.3977	0.1966	0.2607	0.60227
QK-SVM	0.41477	0.41477	0.20734	0.27385	0.58522
H-QCNN	0.41825	0.41923	0.20734	0.27591	0.58174

The results suggest that quantum N-gram language models can improve tweet classification accuracy, especially when using larger N-gram sizes. However, the precision and recall of the models need to be improved to make them more useful in real-world applications.

V. CONCLUSIONS

This paper has presented a new and practical approach to text classification using quantum machine learning models and N-grams. Our experiments show that quantum models can significantly enhance classification performance, especially when combined with higher-order N-grams. This approach has proven improvement in classifying tweets, a notoriously difficult and noisy text classification task. We could attribute this improvement to the probabilistic nature of quantum mechanics that quantum-evaluated models have in common with the N-gram definition. However, we do not yet have evidence for this, and we must conduct further studies to verify this hypothesis. Nevertheless, this study is a significant step forward in applying quantum computing in natural language processing and presents exciting opportunities for future research in this area. As quantum hardware advances, we can expect even more promising results, which could lead to real-world applications in various industries, including social media analysis and cybersecurity.

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